Name: ____________________

MULTIPLICATION,
(BY 10, 100 AND 1000)

Date: ______________

LO: I can multiply whole numbers and decimals by 10, 100 and 1000 by shifting place value.

How does ×10 work?

Multiplying by **10** shifts every digit one column **LEFT**. The digits stay the same; their place value changes.

Common error: 'just add a zero.' That works only for whole numbers. For decimals, you must shift digits (3.4 × 10 = 34, not 3.40).

Examples of place-value shift

	7 × 10 = 70	shift L by 1
	7 × 100 = 700	shift L by 2
	7 × 1000 = 7000	shift L by 3
	0.4 × 10 = 4	decimal shifts too
	3.4 × 10 = 3.40	should be 34

Key idea

Multiplying by 10 = shift digits 1 left. By 100 = shift 2 left. By 1000 = shift 3 left. The decimal point appears to move RIGHT.

3.45 × 100 = 345

Decimal point moves 2 places right

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – whole × 10

Calculate 47 × 10

47 has digits 4 and 7

Shift each one column left

= 470

= 470

Add a zero in the units column to mark the new place.

Example 2 – whole × 100

Calculate 23 × 100

Each digit shifts 2 places left

Add 2 zeros as placeholders

= 2300

= 2300

Multiplying by 100 = multiplying by 10 twice.

Example 3 – decimal × 10

Calculate 3.6 × 10

3.6 has 3 ones and 6 tenths

Shift each one place left

= 36

= 36

The 6 moves from tenths to ones.

Example 4 – decimal × 1000

Calculate 0.045 × 1000

Shift each digit 3 places left

0.045 45.0

= 45

= 45

Decimal point moves 3 places right.

YOUR TURN – PRACTICE (deliberate practice)

1. × 10
Shift digits one column left.

a) 8 × 10 = ___

b) 26 × 10 = ___

c) 0.7 × 10 = ___

d) 3.4 × 10 = ___

2. × 100
Shift digits two columns left.

a) 5 × 100 = ___

b) 32 × 100 = ___

c) 0.06 × 100 = ___

d) 1.45 × 100 = ___

3. × 1000
Shift digits three columns left.

a) 4 × 1000 = ___

b) 12 × 1000 = ___

c) 0.005 × 1000 = ___

d) 2.34 × 1000 = ___

4. Mixed scales
Spot the ×10/100/1000 pattern.

a) 7 × ___ = 700

b) 0.3 × ___ = 30

c) 0.06 × 1000 = ___

d) 8.4 × 100 = ___

5. Two-step
Apply the rule twice.

a) (2.5 × 10) × 100 = ___

b) (0.04 × 1000) × 10 = ___

c) (7 × 100) + (3 × 10) = ___

d) (0.6 × 1000) - (5 × 10) = ___

e) (1.2 × 100) × 10 = ___

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) 3.4 × 10 = 3.40 ☐	b) 0.05 × 100 = 5 ☐	c) 47 × 100 = 4700 ☐	d) ×10 means 'just add a zero' ☐	e) 0.6 × 1000 = 600 ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: __________________	Reason: __________________	Reason: __________________	Reason: __________________	Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A box of pencils contains **100 pencils**. A school orders **34 boxes**. How many pencils in total? Use ×100 to find the answer.

Expression: ______________________________

Simplified: ______________________________

b) A water bottle holds **0.75 litres**. A factory fills **1000 bottles** an hour. How many litres of water are used per hour? Show your working.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?
Did I check my answer using a different method?
Does my answer look reasonable?

WELL DONE!